

ELC -2026 Mid Evaluation Presentation

GROUND RESOLUTION ENHANCEMENT for LEO (Lower Earth Orbit) SATELLITE

Student Team Group - 82

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MOTIVATION

LEO satellite images often have limited ground resolution. High-resolution sensors are expensive and hardware-limited. Software-based enhancement can improve image quality for Earth observation applications.

OBJECTIVE

- The objective of this project is to enhance the ground resolution of LEO satellite images using computer vision techniques like image stacking and super-resolution methods to reconstruct higher-quality images while preserving important spatial details.
- Evaluate the effectiveness of these methods using quality metrics.



fig. Resolution Enhancement Through Stacking

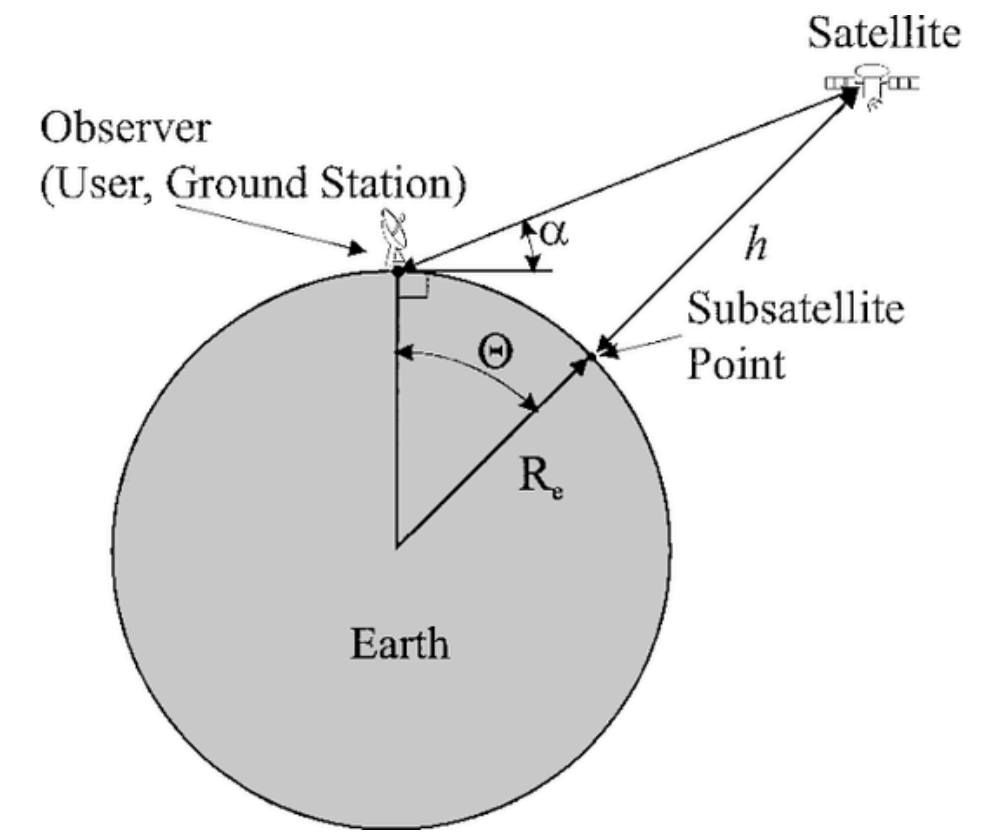
WORK DONE TILL DATE

Swath Analysis

- Studied swath geometry for estimating image coverage.
- Computed swath width using satellite imaging parameters.
- Evaluated the coverage–resolution (GSD) trade-off.
- Integrated swath calculations into the geolocation pipeline.

Latitude & Longitude Estimation

- Studied the GOES-19 ABI geolocation dataset and projection parameters.
- Implemented pixel geolocation using a ray–ellipsoid intersection algorithm from scratch.
- Derived latitude and longitude by solving the viewing-ray intersection with the WGS-84 ellipsoid using satellite geometry.
- Validated computed coordinates against an independent PROJ (pyproj) reference implementation.



Verification Results

```
=====
Pixel          Our Lat   Our Lon   PROJ Lat   PROJ Lon   Error(m)
(11766,783)    -4.545069477117 -150.076789799516 -4.545069477116 -150.076789799515 1.35472228e-07
(6593,11959)   19.908984057678 -84.103454306768  19.908984057675 -84.103454306768 6.16493543e-09
(3094,16848)   41.652717912507 -46.127784126366  41.652717912503 -46.127784126369 2.36901231e-07
=====
```

comparison.csv generated.

Maximum Error : 2.369e-07 metres

Verification PASSED

WORK DONE TILL DATE

1. Studied LEO satellite image acquisition process
2. Implemented various **Image Stacking algorithms (MATLAB)**.
3. Explored image registration and spatial alignment concepts.
4. **Implemented ORB** (Oriented FAST and Rotated BRIEF) robust feature detection and image alignment.
5. Developed a **stacking and stitching pipeline (Python)** for sequential image frames. (Validated the pipeline using real video frame dataset)



Fig. Image Alignmnet + Stacking (Overlap)

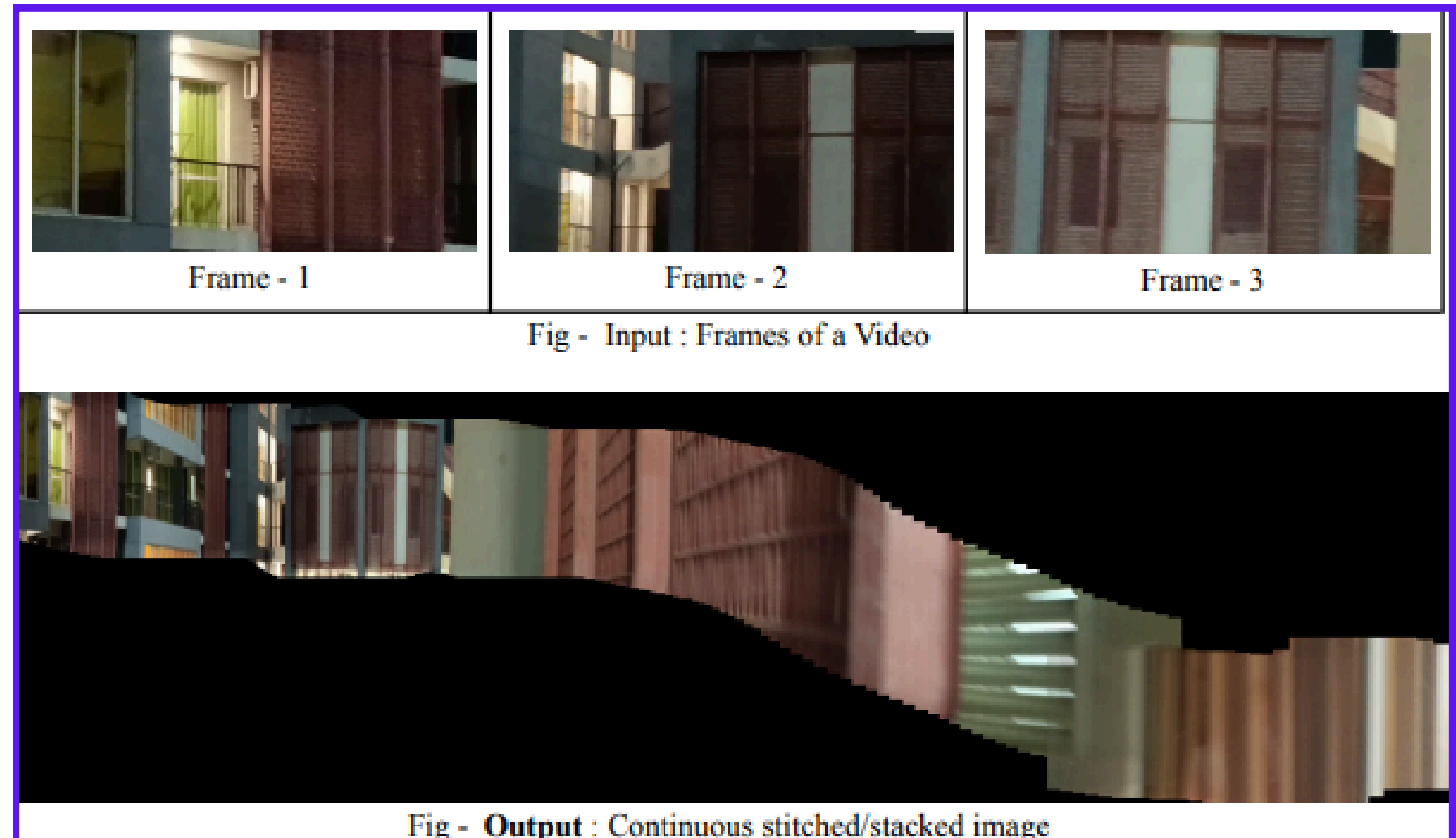
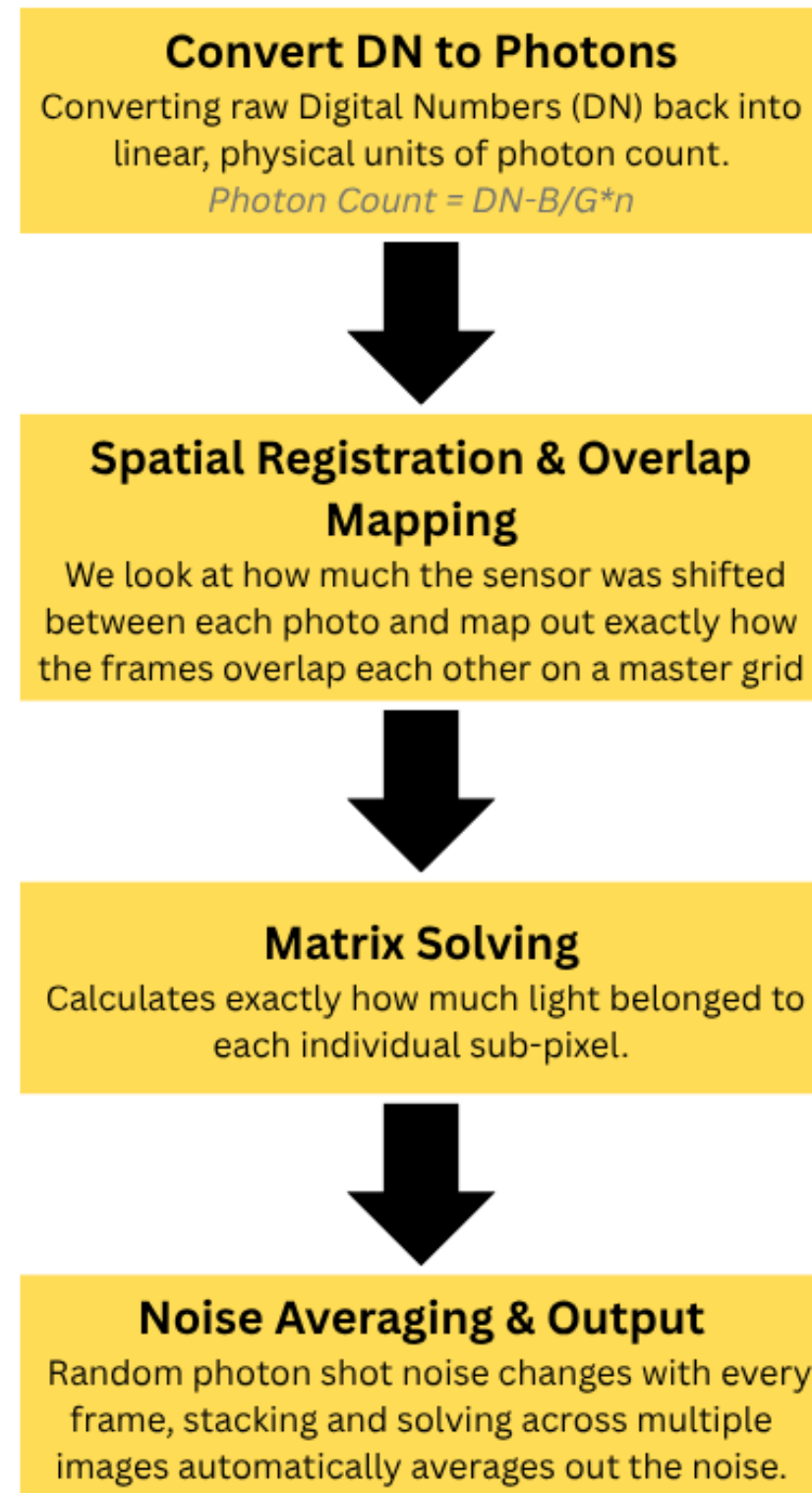


Fig - Input : Frames of a Video

Fig - **Output** : Continuous stitched/stacked image

Multi-Frame Photon Estimation (Sub-Pixel Dithering)



By linearizing our raw camera data into true photon counts, we can treat these overlapping frames as a system of linear equations ($[A]x = b$). A matrix solver then decodes the exact photon intensity of every hidden sub-pixel, physically bypassing the camera's hardware resolution limit.

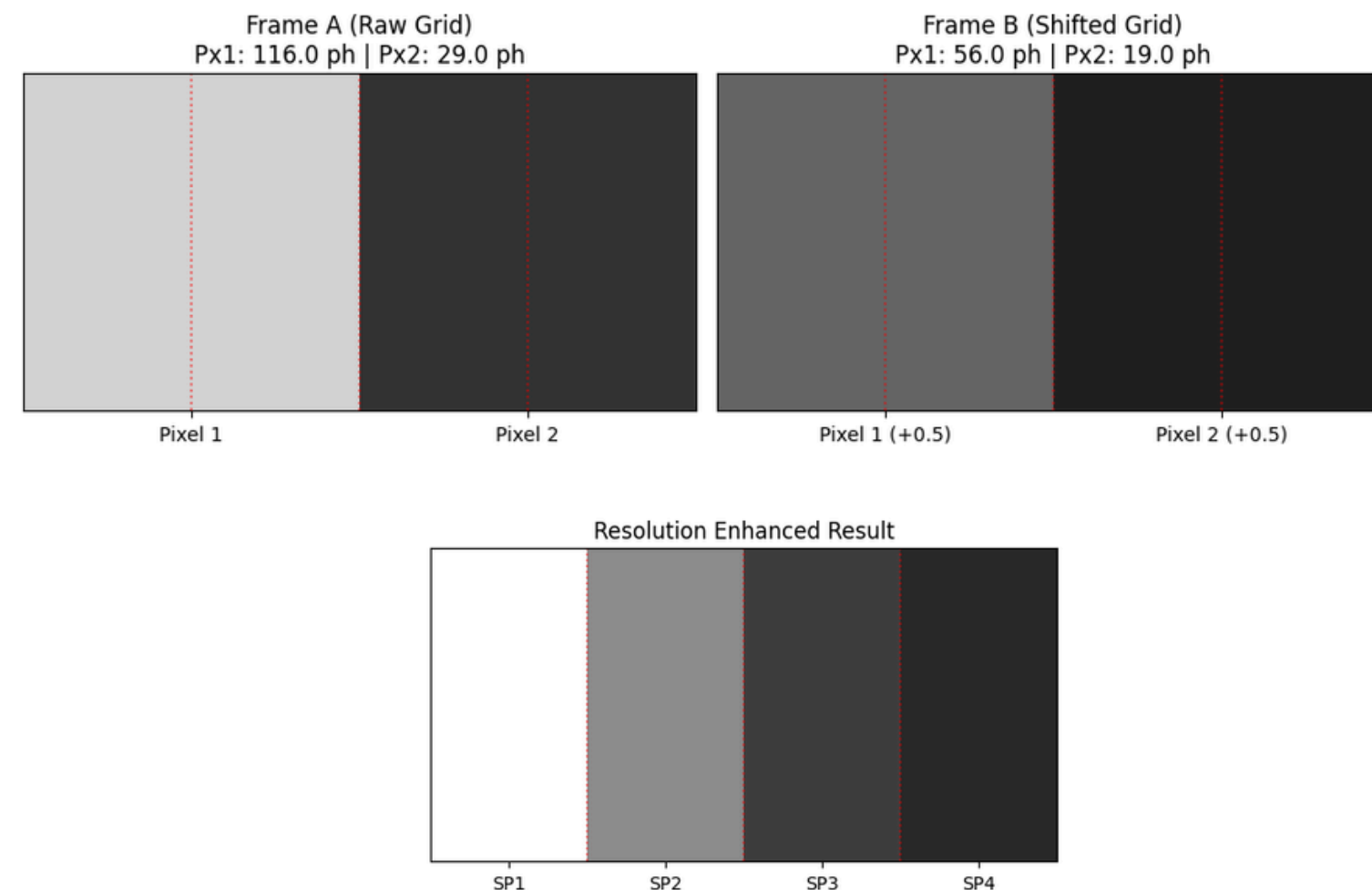


fig. Pipeline for Resolution Enhancement

FUTURE WORK

Resolution Enhancement

- Integrating previous work we will, implementing an image enhancement pipeline using multi-frame photon estimation on real satellite images.
- Evaluating generated images using PSNR and SSIM.

Latitude & Longitude Estimation

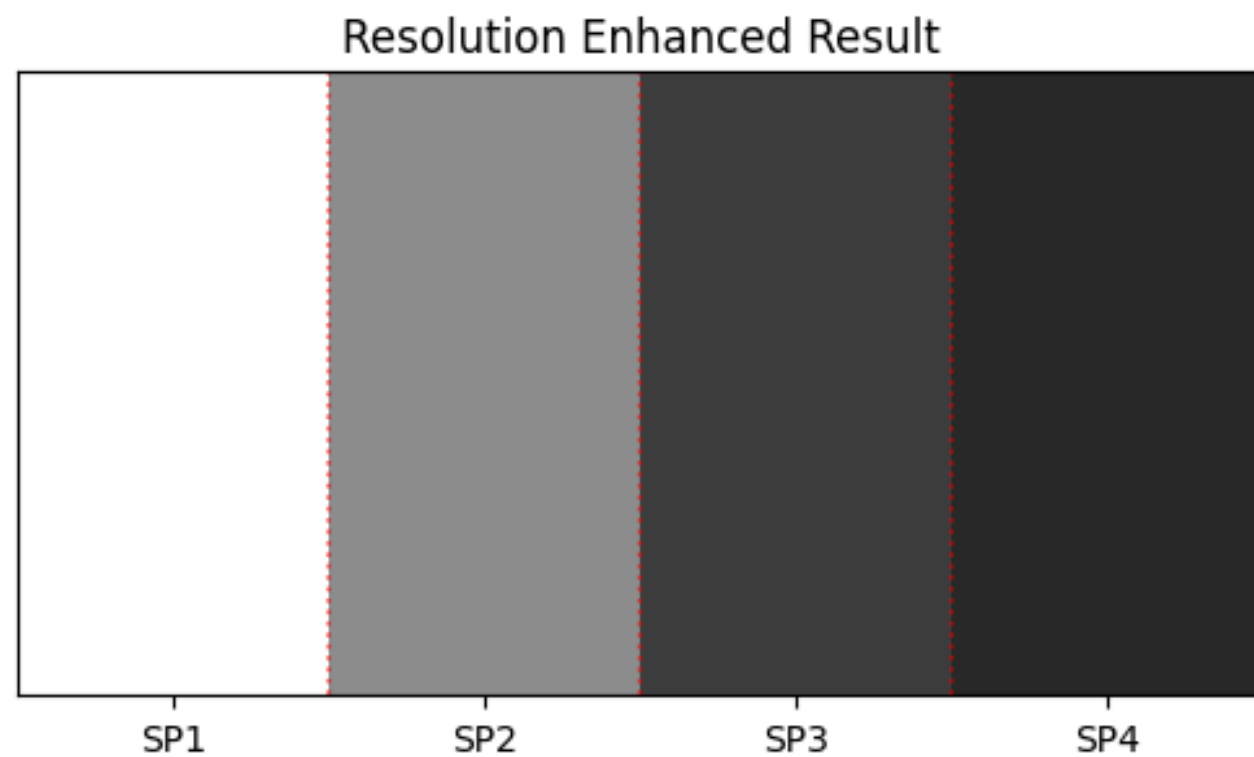
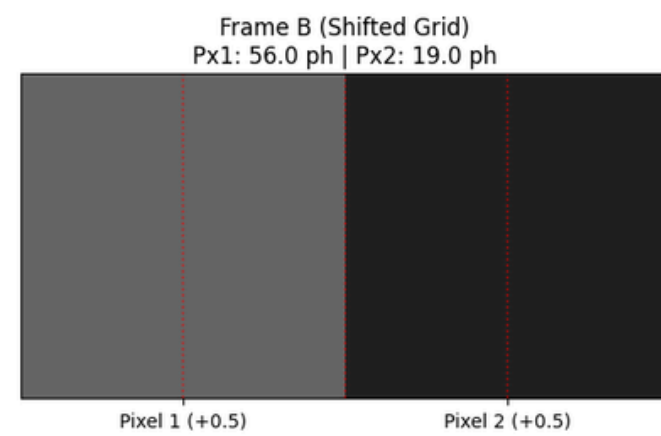
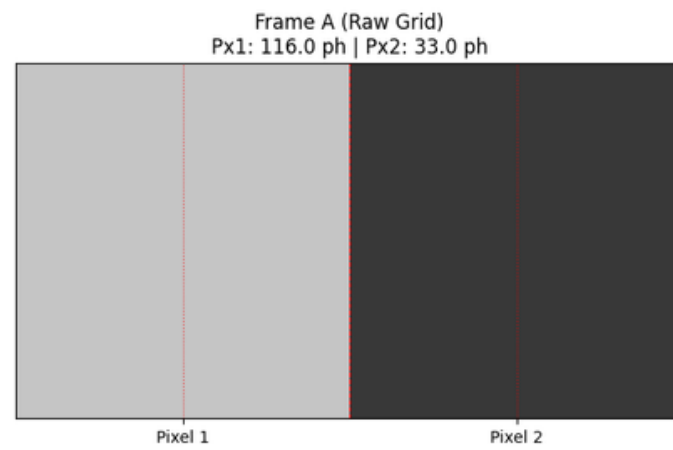
- We will overlay the computed latitude and longitude coordinates onto a reference map to accurately identify the corresponding locations on the Earth's surface.

THANK YOU

All the reports and code generated for this ELC
is available in this GITHUB repository.



https://github.com/ABHINAB-CHHETRI/ELC_Ground-resolution-enhancement-for-LEO-satellite



It sets up a system of equations:

$$1. \quad SP1 + SP2 = 116$$

$$2. \quad SP2 + SP3 = 56$$

$$3. \quad SP3 + SP4 = 33$$

$$[SP1=90] [SP2=26] [SP3=30] [SP4=3]$$